Review Article

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Advances in the management and prevention of chronic pain following breast cancer surgery

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Abstract: Post-mastectomy pain syndrome (PMPS) is a prevalent long-term complication after breast cancer surgery, affecting about 25–40 % of survivors. Its causes are diverse, including surgical injury, adjuvant treatment, and individual vulnerability, yet these insights have not translated into consistent management. Conventional pharmacological therapies provide only modest benefit, while nerve blocks, cryoneurolysis, radiofrequency techniques, and surgical innovations such as autologous fat grafting or nerve reconstruction show encouraging but still limited evidence. Neuromodulation strategies and complementary approaches, including rehabilitation and psychosocial support, further illustrate the diversity of available options. Despite these advances, fewer than half of affected patients receive targeted interventions, and clinical practice varies widely across regions. This narrative review aims to raise clinical awareness of PMPS and to provide an evidence-based reference for practicing clinicians, it summarizes the latest findings on risk factors, preventive measures, and therapeutic options. We contend that PMPS should be recognized as a significant oncological and surgical challenge, warranting systematic management in routine follow-up.

Keywords: post-mastectomy pain syndrome; neuropathic pain; chronic pain; breast cancer; prevention; postoperative management

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Introduction

Post-mastectomy pain syndrome (PMPS) is one of the most frequent but under-recognized complications following breast cancer surgery. It is defined as persistent neuropathic pain in the breast, axilla, chest wall, or inner upper arm lasting longer than three months after surgery, typically manifesting as burning, shooting, electric-shock-like sensations, numbness, or dysesthesia (Figure 1) [1]. Symptoms are frequently exacerbated by physical activity and can coexist with functional limitations and psychological distress [2]. The etiology is multifactorial, encompassing both biological and psychosocial mechanisms. Peripheral nerve injury during mastectomy or axillary dissection, radiation-induced brachial plexopathy, inflammation, and central sensitization are well-established contributors [3]. Additional risk factors include uncontrolled acute postoperative pain, younger age, genetic polymorphisms, and psychosocial distress [4–7].

The prevalence of PMPS varies from 25 to 40 % (Figure 2), reflecting heterogeneity in definitions, populations, and assessment tools. In China, a large single-institution cohort of nearly 2,000 patients reported a prevalence of 28.2 % [1]. In the United States, a prospective tertiary center cohort found that approximately 37 % of women experienced PMPS [8]. European data are similar to findings from the Chinese study, with an Italian prospective study reporting 28 % [9]. Additionally, a study from India reported a prevalence of 35.8 % [10]. Collectively, these studies indicate that PMPS is a common and persistent problem worldwide, regardless of regional differences.

From an oncological perspective, the burden of PMPS extends far beyond pain itself. Breast cancer survival has improved significantly over the past two decades, with five-year survival rates exceeding 90 % overall in many high-income countries [11]. As survivorship increases, the prevalence of long-term treatment sequelae has become a critical determinant of quality of life (QOL). PMPS negatively affects adherence to therapy, delays rehabilitation, and undermines patients' confidence in returning to normal life, thereby reducing the overall effectiveness of cancer care [1]. Despite this burden, intervention rates for PMPS remain strikingly low. In China, only about 10 % of affected patients

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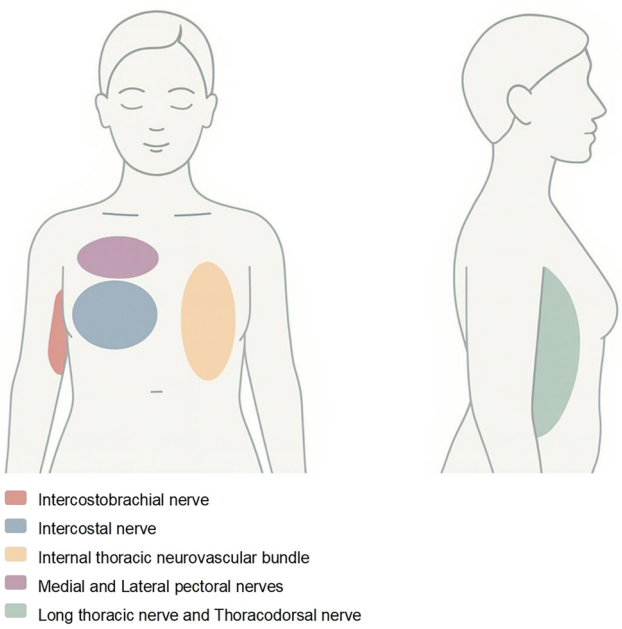

Figure 1: Pain distribution in post-mastectomy pain syndrome (PMPS) and the associated nerve supply.

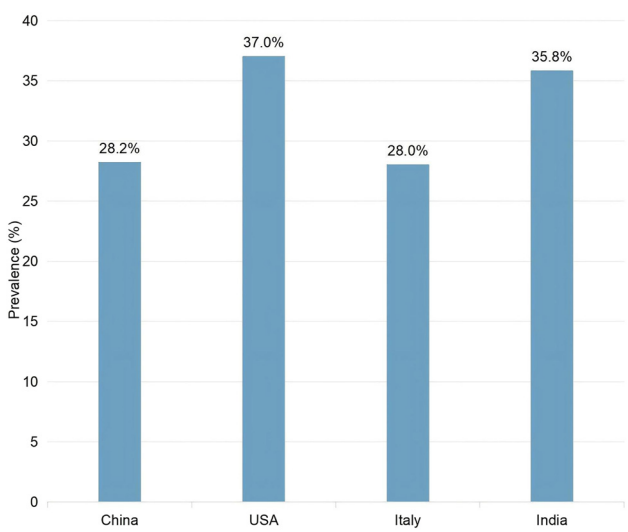

Figure 2: Prevalence of post-mastectomy pain syndrome (PMPS) across representative cohorts. China: Cross-sectional single-institution survey (n=1,983), PMPS defined as pain persisting≥3 months post-surgery and evaluated through telephone interviews, quality-of-life questionnaires, prevalence 28.2 %. USA: Prospective tertiary-center cohort (n=300), assessed≥6 months post-surgery using standardized pain questionnaires, prevalence 37.0 %. Italy: Prospective single-center study (n=307), IASP-based definition, 3-month follow-up with pain scores and interference scales, prevalence 28.0 %. India: Prospective tertiary-center cohort (n=120); pain>3 months in the surgical region, evaluated through pain and quality-of-life questionnaires, prevalence 35.8 %.

received targeted treatment [12]. A Japanese nationwide survey found that merely 47.7 % of physicians provided management, most often with ineffective nonsteroidal anti-inflammatory drugs (NSAIDs) [13]. In India, most women with PMPS chose to endure their pain rather than seek medical intervention [10]. Addressing PMPS is therefore not only a matter of pain control, but an essential component of comprehensive oncological care and survivorship planning.

Several systematic reviews have examined PMPS, focusing mainly on either prevention or treatment [14–16]. However, these studies often addressed isolated aspects of the condition without integrating evidence across its risk factors, preventive strategies, treatments, and rehabilitation approaches. Building upon this foundation, this review provides an updated overview with emphasis on three domains: (1) advances in understanding risk factors; (2) perioperative prevention strategies, including optimized analgesia and surgical refinements; (3) treatment modalities for established PMPS, ranging from pharmacological, interventional, and neuromodulation therapies to surgical innovations and rehabilitation. By synthesizing emerging evidence, we aim to support surgeons, anesthesiologists, and rehabilitation specialists in developing integrated approaches to improve the diagnosis, management, and long-term outcomes of breast cancer survivors affected by PMPS.

This narrative review used a structured search of PubMed and Web of Science for English-language human studies (August 2014–August 2025), with emphasis on 2020–2025 publications. Earlier high-quality studies were cited selectively when recent data were limited. Representative search terms included “postmastectomy pain syndrome,” “neuropathic pain,” “chronic postsurgical pain,” “breast cancer surgery,” and modality-specific keywords such as “nerve block,” “cryoneurolysis,” “radiofrequency,” “targeted muscle reinnervation,” “regenerative peripheral nerve interface,” “autologous fat grafting,” and “rehabilitation.” Given sparse evidence for some emerging modalities, small single-center series and case reports were included and interpreted cautiously. We excluded conference abstracts without full text and non-English publications, and we identified no animal or pediatric PMPS studies.

Risk factors

Demographic factors

Age remains one of the most consistently reported risk factors. Younger patients are significantly more likely to develop PMPS than older patients [4, 12, 17, 18]. High-quality evidence from a meta-analysis demonstrated that the odds of PMPS increased by 36 % for every 10-year decrease in age (OR=1.36, 95 % CI [1.24, 1.48], p<0.001) [4]. This phenomenon may reflect

heightened immune and inflammatory responses, increased central nervous system plasticity, and greater psychosocial distress in younger women. Consistent with this trend, a multicenter cohort study by Ren et al. [12] found that each additional year of age was associated with a 1.8 % reduction in PMPS risk (OR=0.98, 95 % CI [0.97, 1.00]; p<0.001). Notably, they also observed a paradoxical pattern in which patients with more severe pain were older than those with mild pain. This observation may reflect age-related differences in pain perception and coping mechanisms.

A prior history of chronic pain is another important risk factor for PMPS [1, 19]. In one study, patients with pre-existing chronic pain disorders were about twice as likely to develop PMPS (OR=2.15, 95%CI [1.28, 3.06], p=0.021) [1]. This suggests that prolonged pain conditions may induce central sensitization, thereby increasing susceptibility to PMPS. Moreover, reduced educational attainment, inadequate social support, and lack of regular exercise may elevate the risk of chronic pain by impairing coping capacity and psychological resilience [5, 18, 20].

Surgical and treatment-related factors

The extent and duration of surgery are the main surgery-related factors influencing the development of PMPS [1, 21]. Compared with breast-conserving surgery, mastectomy carries a higher risk (OR=2.46, 95 % CI [1.59, 3.94]; p<0.001). In part, this association may be explained by more extensive chest wall dissection and greater disruption of sensory nerves. The risk rises further when axillary lymph-node dissection (ALND) is performed (OR=2.63, 95 % CI [1.58, 4.52]; p<0.001) [1]. Importantly, although the surgical approach strongly affects the likelihood of PMPS, its influence on pain severity appears limited [12].

Radiotherapy is another well-recognized contributor to PMPS. Ren et al. [12] found that the proportion of patients who had received radiotherapy was higher in the PMPS group than in the non-PMPS group (55.3 vs. 41.5 %; p<0.001). Furthermore, radiotherapy was associated with greater pain during movement, potentially linked to treatment-related brachial plexus injury. The research team further found that among patients who underwent radiotherapy, those who developed breast radiation dermatitis (BRD) had a higher prevalence of PMPS (23.4 vs. 13.6 %; p=0.009). Moreover, in those with BRD, earlier onset during radiotherapy was more frequent in the PMPS group (72.5 vs. 59.4 %; p=0.004), and a larger dermatitis area was more common (38.9 vs. 18.4 %; p=0.000) [6]. These associations are compatible with radiation-related neuroinflammation and fibrosis, which can amplify nociception and contribute to chronic pain.

Other controllable factors

Lacking management of postoperative acute pain is identified as a significant predictor for the development of chronic pain. According to Wang et al. [4], each 1-cm increase in postoperative acute pain, measured on a 10 cm visual analog scale (VAS) was associated with a 16 % higher odds of developing PMPS (OR=1.16, 95 % CI [1.03, 1.30]). Moreover, Psychological factors exert a profound influence on the transition from acute to chronic pain [12, 21]. Anxiety, depression, catastrophic thinking, and lack of social support are strongly associated with PMPS development [22].

A recent study suggested that genetic variations may represent a new risk factor for PMPS, although current findings remain preliminary. Vidic et al. [7] found that variants in OPRM1 and miRNA genes may increase susceptibility to PMPS, potentially through alterations in μ-opioid receptor function or miRNA-mediated regulation. In their study of a Slovenian Caucasian cohort, carriers of at least one polymorphic MIR23B rs1011784 allele had higher odds of neuropathic pain (adjusted OR=2.85, 95 % CI [1.07, 7.59]; p=0.036), whereas carriers of two polymorphic MIR107 rs2296616 alleles showed a higher prevalence of chronic pain. Importantly, the allele frequency of OPRM1 has been reported to be considerably higher in Asian populations than in European populations, suggesting that genetic predisposition to PMPS may vary across ethnic groups. Further studies involving larger and more ethnically diverse cohorts, particularly Asian populations, are needed to verify these findings. Although this aspect has received limited attention in previous reviews, it remains a promising but still early area of research that should be interpreted with caution before clinical application.

Prevention strategies

Acute pain management

Nerve block

Effective perioperative analgesia is crucial for PMPS prevention, as poorly controlled acute pain is a major risk factor. Regional anesthesia has been widely applied and investigated as a preventive strategy, and recent studies suggest that its role is gradually extending from alleviating acute pain to potentially influencing long-term outcomes.

Preoperative paravertebral block (PVB) remains one of the most widely used. Parikh et al. [23] showed that PVB lowered pain scores at 2 and 24 h after surgery. Another

randomized controlled trial (RCT) found that a single multilevel PVB (T2–T5) likewise improved acute pain and reduced 6-month PMPS incidence from 24.0 to 12.5 % (RR=0.52, 95 % CI [0.28, 0.96]; p=0.031) [24]. These results suggest that PVB may play a role in preventing both acute and chronic pain. Although the deeper needle insertion carries a potential risk of pneumothorax, neither study reported such complications.

Pectoral nerve block (PECS) and the modified pectoral nerve block (PECS II) are technically simpler and quicker. A study has confirmed that PECS I combined with PECS II, when used with general anesthesia, can significantly improve postoperative analgesia [25]. Furthermore, PECS II has been shown to be non-inferior to PVB in terms of acute analgesia and 24-h morphine consumption [26]. However, evidence from long-term follow-up remains limited. A prospective observational study suggested that PECS II reduced the incidence of PMPS at 3 months (14.9 vs. 31.8 %, p=0.039), though no significant difference at 12 months (2.1 vs. 4.7 %, p=0.654) [27]. A recent single-center RCT also showed that PECS II lowered 12-week PMPS incidence (20.65 vs. 34.78 %, p<0.05) and decreased opioid consumption (p<0.001) [28]. These findings indicate that PECS II may serve as a promising alternative to PVB, given its superficial approach and lower risk of complications, although longer-term, multicenter trials are still required for confirmation.

In addition, the effectiveness and safety of the erector spinae plane block (ESPB) in preventing PMPS have attracted increasing attention. Existing studies have demonstrated that ESPB provides an opioid-sparing effect comparable to PVB [29] and is associated with fewer neuropathic symptoms at 6 months. However, there is still a lack of long-term follow-up data with PMPS incidence as the primary endpoint, and its preventive value remains to be further clarified [30].

Intrathecal morphine

Intrathecal morphine (ITM) is a neuraxial block technique that provides wide-field analgesia with a single injection. Swisher et al. [31] conducted the first study to evaluate the efficacy of ITM in abdominal free flap breast reconstruction. Their findings showed that ITM reduced opioid consumption within 48 h compared with PVB (adjusted difference=−28.86 morphine equivalent dose, 95 % CI [−45.19, −12.52]; p<0.001) and halved the median length of hospital stay (1 vs. 2 days, p<0.001). These results suggest that ITM may represent a more effective strategy than PVB and could potentially reduce PMPS risk, particularly in patients undergoing reconstructions involving both thoracic and abdominal regions. However, this study lacked long-term follow-up data, and its preventive effect remains to be verified.

Transcutaneous electrical acupoint stimulation

Research on non-invasive neuromodulation for PMPS prevention has progressed. In a multicenter RCT of 576 patients, Lu et al. [32] found that preoperative transcutaneous electrical acupoint stimulation (TEAS) at combined acupoints reduced the incidence of PMPS (RR=0.68, 95 % CI [0.52, 0.89], p=0.007) and reduced intraoperative remifentanil use (group difference=−0.20 mg, 95 % CI [−0.34, −0.07]; p<0.001). These findings suggest that TEAS is a promising, low-cost, non-pharmacologic option for PMPS prevention, offering practical advantages over regional blocks or intrathecal techniques.

Surgical technique optimization

ALND is a risk factor for PMPS, potentially due to injury of the intercostobrachial nerve (ICBN). Evidence remains inconsistent on whether preserving the ICBN reduces acute postoperative pain and prevents its progression to PMPS. Andersen et al. [33] previously reported that the preservation, partial preservation, or transection of the ICBN during surgery did not significantly affect the incidence of acute pain within the first postoperative week. In a quantitative sensory assessment study, the team observed lower 1-week movement pain after nerve section. This finding implies that nerve preservation may intensify acute pain as a result of microdamage or mechanical stimulation [34]. By contrast, Kaur et al. [35] found that preserving the ICBN improved sensory outcomes and reduced persistent pain at 3 months (12.5 vs. 31 %, p=0.024) with fewer neuropathic features (8.33 vs. 20.6 %, p=0.004). Consistent with this result, a recent meta-analysis involving 25 RCTs demonstrated that ICBN preservation significantly decreased the PMPS risk (RR=0.27, 95 % CI [0.21, 0.34]) and sensory disturbances (RR=0.25, 95 % CI [0.18, 0.35]) [36]. This study presents the most convincing evidence to date regarding the preventive efficacy of ICBN protection for PMPS, while also considering the potential acute effects of neurostimulation.

Neurotized autologous breast reconstruction may serve as a prophylactic strategy. In an RCT of implant-based breast reconstruction, end-to-end coaptation of the intercostal nerves was performed. The results showed that the coaptation group had a significantly lower Short-Form McGill Pain Questionnaire (SF-MPQ) score at 6 months, suggesting reduced early neuropathic pain. Although the 12-month difference was not statistically significant, a trend toward lower pain persisted in the coaptation group [37]. Interest in nerve reconstruction for breast surgery is increasing, but most studies focus on breast sensory recovery and QOL,

while research using PMPS incidence as the primary study endpoint remains limited [38]. Given the substantial PMPS burden after reconstruction [39], further studies on the preventive role of neurotized autologous reconstruction hold significant clinical relevance and research value.

Summary

In summary, PVB is the most commonly used nerve block for PMPS prevention. PECS II and ESPB are technically simpler approaches that may carry a lower risk of complications, and they show considerable promise. ITM provides broad, uniform analgesia and is particularly relevant for autologous tissue breast reconstruction, while TEAS is a non-invasive, non-pharmacological, and low-cost strategy with encouraging signals, although it is not yet routinely incorporated into clinical pathways. ICBN preservation and nerve coaptation are among the few prevention strategies that surgeons can implement intraoperatively. These approaches provide direct operator control and anatomy-based benefits without dependence on postoperative adherence. Nonetheless, evidence for nerve coaptation remains limited; additional studies are needed to confirm durability, quantify complications, and define appropriate indications. These techniques should be regarded as pragmatic options to integrate with oncologic principles and multimodal analgesia rather than as definitive solutions. For ease of comparison, the key studies on these preventive strategies are summarized in the following table (Table 1).

Treatment strategies

Medication

Pharmacological therapy remains foundational in PMPS management. The Canadian Pain Society endorses gabapentinoids (gabapentin and pregabalin), tricyclic antidepressants, and serotonin noradrenaline reuptake inhibitors as first-line treatments for neuropathic pain [40]. As PMPS is fundamentally a neuropathic pain condition, a double-blind RCT tested preoperative pregabalin 75 mg twice daily for 1 week. Pregabalin reduced 12-week PMPS incidence (RR=0.26, 95 % CI [0.12, 0.56]; p<0.001) and lowered the proportion of patients with neuropathic pain (RR=0.28, 95 % CI [0.14–0.57]; p<0.001) [41].

Topical therapy is a practical option for PMPS, with high-concentration capsaicin patches (HCCP) exhibiting notable clinical efficacy. The International Association for the Study of Pain has incorporated HCCP into its second-line therapy guidelines [42]. Dupoiron et al. [43] administered HCCP to PMPS patients, resulting in 82.3 % of patients attaining at least one significant or complete remission, and an effectiveness rate of up to 70.7 % for early postoperative pain. The team subsequently performed a multicenter randomized controlled experiment to compare the efficacy of HCCP with pregabalin. The findings indicated that a single HCCP therapy was not inferior to daily oral pregabalin in alleviating pain intensity and was more efficacious in diminishing the pain area. The principal adverse effect of HCCP is a localized burning sensation. In contrast to the systemic adverse reactions associated with oral medications, patients exhibit greater tolerance for the former. After two months of treatment, 70.8 % in the HCCP group opted to continue treatment, whereas only 37.3 % in the pregabalin group persisted, and 52.9 % indicated a preference to transition to HCCP [44]. These findings not only validate the efficacy and safety benefits of HCCP in treating PMPS but also indicate patients' preference for topical medicine. HCCP is anticipated to emerge as the primary treatment modality for PMPS in the future.

Interventional nerve techniques

Nerve block

Targeted nerve blocks are an option when pharmacotherapy is inadequate or poorly tolerated. However, compared with the use of nerve blocks for the prevention of PMPS, studies on their therapeutic application are limited. A study performed by the Memorial Sloan Kettering Cancer Center (MSKCC) evaluated 169 patients with PMPS. Using precise anatomical localization, the investigators selected targeted nerve block procedures (Table 2) [45]. The findings indicated a considerable VAS reduction from 7 to 3 (mean difference=3.7 points, 95 % CI [3.58, 3.98]; p=0.0001), with a median duration of pain relief lasting 84 days. However, this approach was not effective for all patients. Nevertheless, approximately 6 % of patients experienced minimal or no benefit [45]. Other small case series have described clinically meaningful relief following ultrasound-guided serratus plane [46–48] or ICBN blocks [49, 50], but follow-up was typically limited to weeks or a few months. Current data therefore indicate that while nerve blocks can achieve significant short-term analgesia, their durability is often modest. Importantly, block response can be used as a diagnostic and stratification tool, identifying patients likely to benefit from subsequent interventions such as radiofrequency ablation, cryoneurolysis, or neuromodulation. In this context, nerve blocks should be regarded less as a

Table 1: Summary of prevention strategies for PMPS.

Prevention strategies	Reference	Evidence level/Design	Patients	Follow-up	Outcomes variables	Main results	Study limitations
PVB	Parikh et al. [23], 2016	Low/Retrospective cohort	78	In-hospital only	Pain scores (VNRS); TTON; LOS; PONV.	VNRS at 2 h: 3 (2–5) PVB vs. 5 (4–6) control, P<0.01; at 24 h: 3 (2–4) vs. 4 (3–5), P<0.01. TTON: 66 (54–77) h vs. 76 (70–94) h, P<0.01). LOS: 95 (89–114) h vs. 116 (95–121) h, P<0.01; 4 (4–5) nights vs. 5 (4–5) nights, P=0.05; and there was no difference in PONV between 2 groups (15 vs. 26 %, P=0.26).	Non-randomized cohort; single center; short follow-up; opioid dosing not collected.
	Lin et al. [24], 2020	Moderate/RCT	218	Acute: To 48 h; chronic: To 12 months	CPSP incidence; pain scores (NRS); neuropathic pain (ID-pain scale); LOS; pain severity and interference; complications.	CPSP at 6 months: 12.5 % PVB vs. 24.0 % control (RR=0.52.95 % CI [0.28, 0.96], P=0.031). CPSP at 12 months: 9.9 vs. 19.4 %, but ns (RR=0.51, 95 % CI [0.25, 1.04]; P=0.055). Neuropathic pain at 6 months: 10.6 vs. 23.1 % (RR=0.46, 95 % CI [0.24, 0.89]; P=0.016); at 12 months:10.9 vs. 22.3 % (RR=0.49, 95 % CI [0.25, 0.95], P=0.028). Acute pain: NRS at rest P=0.006; on movement P<0.001. LOS: No difference (median 7 vs. 6 days; P=0.156). No AEs.	Single-center; no placebo injection in control.
PECS	Bashandy et al. [25], 2015	Moderate/RCT	120	In-hospital only	Pain scores (VAS); intraoperative fentanyl dose; postoperative morphine consumption; need for PCA; PONV score; sedation score; PACU stay; LOS.	PECS group had lower VAS at all time points. Intraoperative fentanyl dose: 115 ± 28.6 μg PECS vs. 252.5 ± 44.4 μg control, P<0.001). Postoperative morphine consumption to keep VAS<3: 2.9 ± 1.71 mg vs. 6.9 ± 1.86 mg, P<0.001; at 0–4 h: 0 (0–9) vs. 4 (2–12) mg, P=0.02 and 4–12 h: 1 (0–4) vs. 4 (0–10) mg, P=0.035. PCA required in 12/60 vs. 36/60. PONV: 0.15 ± 0.37 vs. 1.65 ± 0.88, P<0.001. Sedation: 2.10 ± 0.31 vs. 3.20 ± 0.52, P<0.001. PACU stay: 14 ± 11 vs. 28 ± 12 min, P=0.012. LOS: ≤24 h vs. 12≤24 h, 42≤48 h, 6>48 h; P<0.001.	No sham block; single center; could not assess block quality before induction; PCA not offered to all (triggered only if VAS≥3); short follow-up.
	De et al. [27], 2019	Low/Prospective observational study	140	Acute: To 24 h; chronic: To 12 months	Pain scores (NRS); intraoperative opioids; rescue analgesic requests in 24 h.	24 h mean NRS: 1 (0–3) PECS vs. 3 (1–4) control, P=0.005; maximum NRS 3 (1–4) vs. 4 (2–6), P=0.017. Persistent pain at 3 months: 14.9 vs. 31.8 %, P=0.039; at 12 months: 2.1 vs. 4.7 %, P=0.654. Intraoperative opioid use: fentanyl 1.61 (1.21–2.31) vs. 3.3	Monocentric, non-randomized design; group-size imbalance; surgical procedures were not uniform; analgesic management heterogeneity.

Table 1: (continued)

Prevention strategies	Reference	Evidence level/Design	Patients	Follow-up	Outcomes variables	Main results	Study limitations
						(2.2–4.4) μg/kg/h, P<0.001; remifentanil 0 (0-0) vs. 0.1 (0–0.1) μg/kg/min, P<0.001. Rescue analgesic requests: 14.3 vs. 37.4 %, P=0.006.	
	Hu et al. [28], 2025	Moderate/RCT	98	Acute: To 48 h; chronic: To 12 weeks	PMPS incidence; intraop remifentanil dose; oxycodone dose; time to first analgesia; pain scores (NRS); HADS; median pain duration.	Incidence: 20.65 % PECS vs. 34.78 % control, P<0.05. Intraoperative remifentanil: 1.46 ± 0.11 mg vs. 2.66 ± 0.18 mg, P<0.001. Oxycodone: 22.57 ± 3.21 mg vs. 31.62 ± 4.71 mg, P<0.001. Time to first analgesia: 368.80 ± 157.68 min vs. 96.60 ± 40.12 min, P<0.001. 48 h NRS: 1 (1.1) vs. 4 (3.4), P<0.001, and 12 weeks NRS 1 (1.1) vs. 5 (4.6), P=0.032. PECS reduced HADS (P<0.05) and shortened median pain duration (8 vs. 10 weeks, P<0.05).	Single-center; short follow-up; no weekly pain severity; at 12 weeks, only NRS was collected without full pain phenotype details.
ESPB	Yu et al. [30], 2024	N (protocol with no outcomes)/RCT	160	Acute: To 48 h; chronic: To 12 months	Analgesic consumption; pain scores (VAS); PMPS incidence; inflammatory cytokines; plasma ropivacaine; additional analgesics; recovery metrics; LOS.	Not reported	Single-centre; block operator unblinded.
ITM	Swisher et al. [31], 2025	Low, retrospective cohort	102	In-hospital only	Opioid consumption; pain scores (NRS); LOS; AEs.	Opioids consumption: PACU 0 (0–8.8) ITM vs. 12.5 (0–25) mg PVB, P=0.009; PACU-POD1 0 (0–15) vs. 7.5 (0–30) mg, P=0.046; POD1-POD2 7.5 (0–22.5) vs. 30 (10–46.3) mg, P=0.002. Opioids through POD2: −28.86 MEQ vs. PVB (95 % CI [−45.19, −12.52], P<0.001). NRS: PACU 0 (0–2) vs. 2 (0–4), P=0.015; PACU-POD1 0.5 (0–2.5) vs. 3 (0–6), P=0.001; POD1-POD2 3 (0–5) vs. 4 (2–6), P=0.09 (ns). LOS: 1 (1–2) day vs. 2 (2–3) days, P<0.001. AEs rates were both 39 %. In the ITM group were hypotension/orthostasis (33.3 %), plus pruritus (3.9 %), nausea (2.0 %), possible PDPH (2.0 %), and delayed emergence (2.0 %).	Retrospective design; non-randomized; potential changes in multimodal regimen over time; incomplete standardization of abdominal blocks in controls; short follow-up.
TEAS	Lu et al. [32], 2021	Moderate, RCT	576	6 months	Incidence of PMPS; intraop remifentanil; PONV; pain scores (NRS); rescue analgesic and satisfaction.	Combined-acupoint TEAS reduced 6 months PMPS incidence vs. sham (22.1 vs. 34.6 %, RR=0.68, 95 % CI [0.52, 0.89], P=0.07) and vs. single-acupoint (22.1 vs. 36.4 %, RR=0.72, 95 %	Patient blinding not feasible; age>65 excluded; no non-acupoint stimulation comparator in this trial (authors cite prior

Table 1: (continued)

Prevention strategies	Reference	Evidence level/Design	Patients	Follow-up	Outcomes variables	Main results	Study limitations
						CI [0.55, 0.93], P=0.012). Intraop remifentanil: −0.21 mg, 95 % CI [−0.34, −0.07], P<0.001 vs. sham; ns vs. single. PONV: Combined vs. single RR=0.71, 95 % CI [0.53, 0.94], P=0.004; vs. sham RR=0.60, 95 % CI [0.45, 0.81], P<0.001. NRS/rescue analgesic and satisfaction: ns.	work); mechanism not directly measured.
ICBN Preservation	Kaur et al. [35],2021	Low/Prospective randomized study	53	90 days	Sensation test; pain scores (NRS); neuropathic pain (DN4 questionnaire).	Sensory loss (12.5 % in ICBN preservation group vs. 41.37 % in control, P=0.020). Pain incidence (12.5 vs. 31 %, P=0.024). Pain scores (1.54 ± 0.65 vs. 2.27 ± 0.95, P=0.003). Neuropathic pain incidence (8.33 vs. 20.6 %, P=0.004).	Small single-center study; short follow-up; convenience sample; surgeons were not blinded.
	Li et al. [36], 2025	Moderate/ Meta-analysis	2,560	≥12 months	CPSP incidence, pain intensity (NRS); sensory disturbance; cancer recurrence; duration of surgery; LOS; blood loss; AEs.	Moderate certainty: ICBN preservation probably reduces CPSP (RR=0.27, 95 % CI [0.21, 0.34]) and sensory deficits (RR=0.25, 95 % CI [0.18–0.35]); low certainty: may reduce pain intensity (WMD −0.55 cm on 10-cm VAS, 95 % CI [−0.68 to −0.42]); moderate-to-severe CPSP (RR=0.26, 95 % CI [0.10–0.65]); hypoesthesia (RR=0.40, 95 % CI [0.25–0.64]) and paraesthesia (RR=0.30, 95 % CI [0.13–0.69]); likely increases operative time (mean +11 min, 95 % CI [4, 18]), while LOS (WMD −0.21 days, −0.79 to +0.38) and blood loss (WMD +1.66 mL, −0.72 to +4.04) show no or little difference. No AEs.	Most trials had a high risk of bias; incomplete reporting of adjuvant therapies; inconsistent definitions of sensory outcomes; the evidence base is dominated by Chinese RCTs, with potential; in some studies, the CPSP definition did not adequately exclude other causes; and the evidence on recurrence risk is uncertain.
Intercostal nerve coaptation	Kim et al. [37], 2025	Low to moderate/RCT	15	12months	SF-MPQ; pain scores (NRS); PPI; BREAST-Q (PWBC).	At 6 months, the coaptation group had a lower SF-MPQ (3.25 ± 1.39 vs. 5.57 ± 2.44, P=0.038); NRS, PPI, and BREAST-Q showed no differences (P=0.772, 0.483, 0.768). At 12 months, SF-MPQ, NRS, PPI, and BREAST-Q all showed no differences (P=0.183, 0.704, 0.738, and 0.436).	Small sample; No quantitative measure of neuroma formation possible.

VNRS, verbal numeric rating scale; TTON, timeto oral-only narcotics; LOS, length of stay; PONV, postoperative nausea and vomiting; CPSP, chronic postsurgical pain; NRS, numerical rating scale; AE, adverse event; PCA, patient-controlled analgesia; MEQ, morphine equivalent; PACU, post-anesthesia care unit; HADS, hospital anxiety and depression scale; RR, risk ratio; CI, confidence interval; ns, not significant; POD, post-operative day; PDPH, post-dural puncture headache; WMD, weighted mean difference; SF-MPQ, short-form McGill pain questionnaire; PPI, present pain intensity; BREAST-Q (PWBC), the breast-Q reconstruction module-physical well-being: chest.

Table 2: Pain regions and nerve block techniques (summarized from Yang et al. [45], 2021).

Pain region	Nerve block technique	Injection site
Medial arm with or without axilla	Deep serratus plane block	At the 2nd or 3rd rib, anterior to the mid-axillary line
Anterolateral chest wall or deep breast pain	Deep serratus plane block (ESPB for diffuse pain)	At the 4th or 5th rib at the mid-axillary line (over the corresponding rib, medial to lateral technique for ESPB)
Axillary or midaxillary chest wall pain	Superficial serratus plane block or long thoracic/thoracodorsal nerve block	At the 4th rib anterior to the midaxillary line or between the latissimus dorsi and serratus anterior muscles
Midline chest wall or sternal pain	Parasternal block (also referred to as the transversus thoracis muscle plane block)	Lateral to the sternum, between the 2nd and 4th intercostal spaces (at the rib and sternum junction)
Upper anterior chest wall pain (above the breast)	PEC I block or PEC II block	Between pectoralis major and minor muscles (PEC I) or between pectoralis minor and serratus anterior at the 3rd/4th ribs (PEC II)
Isolated dermatomal distribution pain	Intercostal nerve block or paravertebral block	At the angle of the rib (intercostal) or paravertebral space
Diffuse ipsilateral arm pain (with CRPS features)	Stellate ganglion block Or T2 thoracic sympathetic block	Stellate ganglion or T2 paravertebral region
Diffuse ipsilateral chest wall pain (with CRPS features) or phantom breast pain	T4 or T5 paravertebral block	T4 or T5 paravertebral space

ESPB, erector spinae plane block; PEC, pectoral nerve block; CRPS, complex regional pain syndrome.

definitive therapy and more as the first step in a stepwise interventional strategy.

Radiofrequency ablation

Radiofrequency ablation (RFA) can complement diagnostic nerve blocks. Nerve blocks not only provide rapid analgesia but also help localize the pain generator. For patients who respond well to diagnostic blocks, subsequent RFA may achieve longer and more sustained pain relief [51–53]. Commonly used RFA techniques include thermal RFA, pulsed radiofrequency (PRF), and cooled RFA (C-RFA) [54]. PRF is a non-destructive neuromodulation technique. Fam et al. were the first to evaluate the efficacy and safety of PRF combined with corticosteroid injection applied to the dorsal root ganglion (DRG) for the treatment of PMPS. In patients with refractory PMPS attributable to ICBN injury, significant improvements were observed after 6 months, including a reduction in VAS from 7.48 to 4.7, an increase in QOL scores from 4.66 to 6.88, and over 64 % of patients expressed willingness to undergo the treatment again [55]. The DRG is a cluster of primary sensory neuron cell bodies and serves as a critical hub in nociceptive transmission. PRF targeting the DRG allows precise modulation of neuronal excitability. However, the procedure is technically demanding and carries risks such as pneumothorax and vascular injury [56]. The paravertebral nerve (PVN), representing the proximal spinal nerve segment emerging from the DRG, offers greater technical feasibility and safety as an interventional target. Hetta et al. [57] compared PRF applied to the DRG vs. the thoracic PVN. Both groups achieved comparable short-term analgesia, but at 6 months, 75.9 % of DRG-treated patients attained over 50 % pain relief compared with only 24.1 % in the PVN group. Given the complexity and potential risks associated with DRG interventions, the investigators recommended PVN as the first-line PRF target for PMPS, reserving DRG as a second-line option when PVN PRF fails.

The stellate ganglion (SG) is a sympathetic ganglion that supplies the upper limb, chest wall, and upper thorax [58]. In some patients with PMPS, abnormal sympathetic sensory coupling is thought to sustain pain, making the SG an important target for interrupting sympathetically maintained pain. Building on this mechanism, Abbas et al. [59] evaluated the efficacy and safety of thermal RF at the SG in PMPS. At 6 months, 54.3 % of patients receiving thermal RF achieved over 50 % pain relief compared with 30.5 % with pulsed RF. Moreover, the thermal RF group also showed greater improvements in pain intensity and reduced opioid consumption. More recently, another study extended the sympathetic targets by applying thermal RF to the thoracic sympathetic ganglia (T2–T4). All patients underwent diagnostic thoracic sympathetic blocks before enrollment to verify sympathetically maintained pain. 83.3 % of patients achieved pain relief, with consistently lower pain intensity across follow-up. In addition, among those with breast cancer–related lymphedema (BCRL), improvement was observed in 72.7 % of cases [52]. By incorporating sham controls and rigorous patient selection, this study provided evidence supporting the potential clinical benefit of thoracic

sympathetic RF ablation in patients with sympathetically maintained PMPS.

Cryoneurolysis

Cryoneurolysis is a percutaneous analgesic technique that rapidly cools the target nerve to extremely low temperatures, inducing wallerian degeneration and reversible axonal injury, thereby interrupting nociceptive signal transmission [60]. In recent years, this technique has gained increasing attention in the management of PMPS. Ilfeld et al. [61] performed preoperative, ultrasound-guided cryoneurolysis as a preventive intervention. The results demonstrated not only a marked reduction in acute postoperative pain (postoperative day 2 median difference=−2.5, 97.5 % CI [−3.5, −1.5]; p<0.001), but also decreased the incidence of PMPS from 17 to 3 %. Nezami et al. [62] later used CT-guided cryoneurolysis targeting the ICBN in established PMPS and also observed pain reduction, with mean pain decreasing by 2.9 points from baseline (95 % CI [−4.6, −1.2]; p=0.0028). A recent review synthesized the existing evidence and suggested that cryoneurolysis provides a clinically meaningful reduction in PMPS-related pain (Hedges' g=−0.963, 95 % CI [−1.55, −0.373]). However, effect estimates varied across studies (I^2=70 %), likely reflecting differences in anatomic targets, imaging guidance, treatment timing, and procedural parameters, as well as a limited overall evidence base [63]. Overall, cryoneurolysis is minimally invasive and can provide meaningful benefit from a single treatment session. It may serve as a promising step-up option for patients with PMPS who initially respond to nerve blocks but later develop recurrent or persistent pain. Nonetheless, larger and more rigorous studies are still required to confirm its efficacy and safety.

Neuromodulation techniques

Unlike interventional procedures that rely on nerve blockade or ablation, neuromodulation techniques offer reversible regulation of pain pathways without permanent tissue damage. They allow repeated application, individualized adjustment, and modulation across peripheral, spinal, and cortical levels, making them particularly valuable for refractory PMPS.

Spinal cord stimulation (SCS) involves epidural implantation of electrodes to deliver electrical impulses to the thoracic dorsal columns, thereby modulating nociceptive transmission. In a retrospective study of 78 women with refractory PMPS, 78.2 % achieved meaningful pain relief (predefined as ≥4-point VAS reduction at 6 months), with mean VAS improving from 7.87 to 3.72. On multivariable analysis, younger age (HR=0.882, 95 % CI [0.802, 0.970], p=0.009), lower BMI (HR=0.659, 95 % CI [0.487, 0.891], p=0.007), shorter pain duration (HR=0.696, 95 % CI [0.505, 0.959], p=0.027), and psychological treatment (HR=0.015, 95 % CI [0.001, 0.377], p=0.011) were independent predictors of favorable outcomes, whereas prior radiotherapy was associated with reduced efficacy [64]. These findings underscore the potential of SCS for patients with widespread, refractory neuropathic pain, while highlighting the need for careful patient selection and prospective validation.

Percutaneous electrical nerve stimulation (PENS) is a minimally invasive form of peripheral neuromodulation technique that uses percutaneously inserted needle electrodes to stimulate target nerves. Wong et al. [65] described a single case of serratus plane PENS in a patient with long-standing, treatment-resistant PMPS who had previously experienced only transient relief from diagnostic nerve blocks. Following a 25-min stimulation protocol, the patient's pain score decreased from 7 to 3, and upper limb function markedly improved, with the benefits persisting for 6 months. Although based on a single case, this report suggests that PENS may help prolong the analgesic effects of diagnostic nerve blocks in appropriately selected patients.

High-frequency rTMS targeting the contralateral primary motor cortex (M1) has been investigated as a non-invasive approach to modulate central pain networks. Kataria et al. [66] conducted a double-blind, sham-controlled study in PMPS and found that rTMS significantly reduced pain, improved QOL and enhanced thermal sensory thresholds on quantitative sensory testing. Treatment was well tolerated, with no withdrawals due to adverse effects. While follow-up was limited to the three-week intervention period, these findings support rTMS as a safe, non-invasive neuromodulation option for PMPS that warrants longer-term, larger studies to confirm durability and refine patient selection.

These studies suggest that neuromodulation may extend the therapeutic spectrum for PMPS by achieving not only pain reduction but also improvements in functional recovery, QOL, and sensory thresholds. Nevertheless, current evidence remains preliminary: follow-up periods are short, sample sizes are modest, and the level of evidence ranges from single case reports to small randomized trials. Future research should focus on conducting large-scale, multicenter trials with longer follow-up, establishing standardized stimulation protocols, and defining patient selection criteria

to clarify which subgroups derive the greatest benefit from each neuromodulation modality.

Surgical treatments

Autologous fat grafting

In the surgical management of PMPS, autologous fat grafting (AFG) has emerged as an increasingly discussed and investigated option. Beyond improving scar pliability and soft tissue quality, recent studies suggest that AFG may also confer meaningful analgesic benefits, with many patients reporting reduced pain intensity and decreased reliance on analgesic medication [67, 68]. These dual functional and aesthetic advantages distinguish AFG from more destructive surgical or interventional procedures and contribute to its growing popularity among surgeons.

AFG was first applied to PMPS patients by Caviggioli et al. [69] who enrolled 113 women after mastectomy and ALND. Among the 72 patients who underwent AFG and were followed for 13 months, 82.4 % discontinued analgesic medication, and pain scores decreased significantly (3.23 ± 2.96 vs. 1.04 ± 2.71, p=0.0005). Subsequent studies extended the technique to patients after breast-conserving surgery and confirmed favorable analgesic effects as well [70]. In a multicenter prospective trial, Lisa et al. [71] evaluated scar remodeling using the Patient and Observer Scar Assessment Scale (POSAS). Pain scores declined substantially at 6 months, accompanied by marked improvements in scar-related parameters. Multivariate analysis indicated that smoking and ALND were negative predictors of clinical response.

However, the current evidence base remains limited. Most published studies are small, heterogeneous in methodology, leaving uncertainty about the durability of pain relief and long-term recurrence rates. Moreover, the optimal technical parameters, such as the amount and frequency of grafting, patient selection criteria, and the role of adjunctive processing methods, have yet to be standardized. While oncologic safety data from breast reconstruction literature are generally reassuring, continued surveillance is warranted in breast cancer survivors.

Nerve reconstruction techniques

Targeted muscle reinnervation (TMR) has recently emerged as a surgical therapeutic option. O'Brien et al. [72] suggested reattaching injured intercostal nerve stumps to nearby motor nerve branches (such as the serratus anterior, adjacent intercostal muscles, pectoralis minor, or pectoralis major nerves). Participants' QOL and postoperative discomfort were evaluated using the BREAST-Q scale, with a mean score of 77.5 approximating healthy norms. Despite suggesting improvements in PMPS-related outcomes, the efficacy of TMR remains constrained by the small sample and lack of controls. The authors note that a prospective comparative study using more specific pain metrics is currently underway.

Advances in surgical neuromodulation have also introduced the regenerative peripheral nerve interface (RPNI), a nerve–muscle interface reconstruction technique with proven efficacy in amputation and limb surgeries [73]. Hart et al. [74] reported that intercostal sensory neurectomy combined with RPNI demonstrated promising preliminary efficacy in the management of PMPS. The postoperative VAS pain score decreased from 9 to 1, and 86 % of patients experienced substantial or complete pain relief. The authors concluded that this technique is safe, technically straightforward, and effective in alleviating PMPS, suggesting it as a potential surgical option warranting further investigation and validation. Recently, Nolan et al. [75] performed prophylactic intercostal RPNI during mastectomy with breast reconstruction. After a median follow-up of 19 months, only 5.3 % of patients developed PMPS.

Compared with TMR, RPNI involves direct reinnervation of muscle grafts rather than motor nerve transfer and is technically simpler. Nevertheless, both studies reported a small number of postoperative complications such as cellulitis and seroma, underscoring the need for future research to determine whether these events are related to surgical technique and to better quantify the overall safety profile of RPNI.

Comprehensive rehabilitation treatment

With the suppression of inflammatory reactions, moderate physical exercise and motor activity may help reduce PMPS symptoms. Using the International Physical Activity Questionnaire, Calapai et al. [76] evaluated the physical activity levels of patients with breast cancer following surgery and examined the correlation between physical activity levels and inflammatory markers and PMPS. Compared to the sedentary group (metabolic equivalent<700), the physically active group (metabolic equivalent≥700) experienced a 32 % reduction in

pain scores and a negative correlation with inflammatory marker levels. Future studies aimed at creating focused exercise therapies to decrease inflammation and alleviate PMPS can theoretically be based on this finding.

Young patients with breast cancer may experience chronic pain that is made worse by anxiety and sleep issues that arise soon after surgery. Pain treatment can be enhanced by performing psychological evaluations on patients, recognizing negative emotions or catastrophic thinking quickly, and putting focused psychological interventions into practice. The efficacy of mindfulness-based cognitive therapy in managing chronic pain following breast cancer surgery was assessed in an RCT. According to the findings, mindfulness-based cognitive treatment improved QOL, decreased the need for analgesics, and significantly decreased pain intensity. These effects persisted over a 6-month follow-up [77].

According to preliminary research, laser acupuncture yielded immediate symptom relief at the first session for most patients. In one study, 67.9 % of patients reported that pain on pressing improved immediately after the first treatment. With follow-up, pain scores decreased from 6.86 to 1.29 (95 % CI [4.90, 6.24]; p<0.001). Static mechanical allodynia declined from 6.77 to 1.75 (95 % CI [4.09, 5.96]; p<0.001) [78] Laser acupuncture is a non-invasive physical rehabilitation technique. It is a viable supplementary treatment for PMPS due to its non-invasive nature and quick alleviation effects, but higher-level data is required to justify its usage.

Summary

Across available options, pharmacotherapy remains the most accessible adopted option. Oral agents generally yield

Table 3: Summary of treatment strategies for post-mastectomy pain syndrome (PMPS).

Treatment strategies	Reference	Evidence level/ Design	Patients	Follow-up	Outcomes variables	Main results	Study limitations
Pregabalin	Reyad et al. [41], 2019	Moderate/RCT	200	24 weeks	PMPS incidence; neuropathic pain (GSNP); pain intensity (VAS); analgesic consumption; safety.	12-week PMPS incidence: 11 % in pregabalin vs. 29 % in placebo, RR=0.26, 95 % CI [0.12, 0.56], P<0.001. Prevalence of neuropathic pain: 13 vs. 35 %, RR=0.28, 95 % CI [0.14, 0.57], P<0.001. VAS: 27 ± 18 vs. 35 ± 21, P=0.002. Analgesic consumption: No differences. Safety was comparable: 7 vs. 5 %, P=0.552, with transient events such as nausea, vomiting, sedation, dizziness, myalgia, myoclonus, asterixis, and blurred vision.	Single-center; short follow-up; loss to follow-up (n=19/200).
HCCP	Dupoiron et al. [43], 2022	Low/Real-world observational study	279	Not reported	CGIC; pain aetiology, pain duration; post-surgical PNP; safety.	CGIC: Complete/important effect 82.3 % per patient, complete/important effect 63.9 % per application. Pain aetiology: Post-surgical pain 63.9 %. Post-surgical PNP (<12 months from onset): 70.7 % of applications produced important/complete relief; even>10 years after onset, response was 56.0 %. Among first-application non-responders, 63.6 % later achieved mild–important/ complete relief with	Single-center observational design; no *a priori* sample-size calculation; primarily descriptive analyses with exploratory subgroup comparisons.

Table 3: (continued)

Treatment strategies	Reference	Evidence level/ Design	Patients	Follow-up	Outcomes variables	Main results	Study limitations
						additional applications. Local reactions in 54.4 % (burning/ pain 45.9 %, erythema 30.8 %); high blood pressure 7.2 %; other systemic AEs were rare.	
	Dupoiron et al. [44], 2025	Moderate/RCT	140	2 months	Pain intensity (NPRS); PRO (PGIC); patient preferences; change in painful area; AEs.	NPRS: −1.926 in HCCP vs. −1.634 in pregabalin, 90 % CI −0.889 – +0.260. PRO: 68.7 vs. 68.9 %, higher missingness in pregabalin. No patients switched from HCCP to pregabalin, whereas 27/51 in the pregabalin arm wished to switch to HCCP. Painful area: mean, 66.1 vs. 91.9 cm^2, P=0.0205; median, 56.6 vs. 80.7 cm^2, P=0.0471. No serious AEs.	Underpowered sample; open-label design; crossover and missing data issues; short follow-up; multiplicity not controlled.
Nerve block	Yang et al. [45], 2021	Low/Retrospective study	169	6 months	Pain intensity (VAS); duration of relief; analgesic consumption; injectate dose comparisons; safety.	VAS fell 56 %, from 6.7 to 3.0 (mean difference 3.7 points, 95 % CI [3.58, 3.98], P=0.0001). Duration of relief (responders): Mean 45 days (median 84). More than 69 % patients achieved at least 60-day relief. Analgesic consumption decreased by 11 %, t=0.72, P=0.47 (ns). Dose effect (triamcinolone): 40 vs. 20 mg yielded an additional about 11 % pain reduction and 17 days longer relief. No AEs.	Single-center retrospective; non-uniform follow-up; multimodal concomitant meds; limited secondary outcomes; no matched control cohort.
PRF	Fam et al. [55], 2018	Low/clinical trial	100	6 months	Pain intensity (VAS); QOLS; analgesic consumption; patient satisfaction; safety.	VAS fell from 7.48 ± 1.46 to 4.70 ± 2.88, author-reported t-tests vs. hypothesized mean 5.5: t = −2.579, P=0.005712). QOLS rose from 4.66 ± 1.95 to 6.88 ± 2.40, t=5.683, P<0.00001. Analgesic use showed a transient reduction at 1 month with no significant long-term reduction. Patient satisfaction: 64.68 % said they would "certainly" repeat PRF and 66.64 % would "certainly" recommend it to other people. No serious complications.	Single-arm, no control; single center; limited sample; short follow-up; PRF parameters not definitive.
	Hetta et al. [57], 2020	Moderate/RCT	64	6 months	Pain relief≥50 %; pain intensity (VAS); analgesic discontinuation;	Pain relief≥50 %: 75.9 % DRG vs. 24.1 % PVN, P<0.001. VAS: 3.0 ± 1.94 vs. 5.1 ± 1.5 (mean difference=1.9, 95 % CI [1.0,	Short follow-up; non-uniform analgesic regimens; concomitant pain conditions (some

Table 3: (continued)

Treatment strategies	Reference	Evidence level/ Design	Patients	Follow-up	Outcomes variables	Main results	Study limitations
					GPE; procedure time; AEs.	2.9], P<0.001). Analgesic discontinuation: 51.9 vs. 6.9 %. GPE: 6 (5–7) vs. 3 (2–4), P<0.001. Procedure time: 27.9 ± 6.4 min vs. 19.1 ± 3.7 min, P<0.001. No major neurological or pulmonary complications.	patients also received analgesics for other pain disorders); no adjunct steroid injection.
Thermal RF	Abbas et al. [59], 2018	Moderate/RCT	80	24 weeks	Responder rate (≥50 % pain relief); pain intensity (VAS); analgesic consumption; QOL; safety.	Responder rate: 54.3 % in thermal RF vs. 30.5 % in PRF, p<0.001. Pain intensity: 31.3 ± 6.9 vs. 65.4 ± 7.6, Δ=45.2 ± 5 vs. 11.4 ± 6.2, P<0.001. Thermal RF produced a larger reduction in analgesic consumption (P<0.05). QOL: No differences between groups (P>0.6). 2 patients in thermal RF developed transient upper eyelid ptosis (Horner's syndrome).	Single-center; short follow-up.
	Hetta et al. [52], 2021	Moderate/RCT	66	6 months	Responder rate (≥50 % pain relief); pain intensity (VAS); analgesics consumption; GPE; BCRL; safety.	Responder rate: 83.3 % in thermal RF vs. 58.0 % in sham, P=0.032. Pain intensity: 2.07 ± 1.53 vs. 2.90 ± 1.15, Δ = −0.87 (95 % CI [−1.56, −0.18], P=0.015). No analgesic use: 40 vs. 0 %, P=0.001. GPE: 7 (5–7) vs. 5 (4–6), P<0.001. BCRL:72.7 vs. 20 %, P=0.018. No complications.	Short follow-up; QOL not directly measured; no local anesthetic/steroid co-injection; no objective anhidrosis assessment.
Cryoneurolysis	Ilfeld et al. [61], 2022	Moderate/RCT	60	12 months	POD2 average pain (NRS); opioid consumption (first 3 weeks); PMPS incidence; safety.	POD2 average pain: 0 (0–1.4) cryoneurolysis vs. 3.0 (2.0–5.0) sham, difference −2.5 (97.5 % CI −3.5 to −1.5, P<0.001). Opioids consumption: 1.5 (0–14) mg vs. 72 (20–120) mg, P<0.001. PMPS incidence: 3 vs. 17 %, P<0.001. No major complications.	Single center; bundled analgesia may influence generalizability; provider unblinded for technical reasons.
	Nezami et al. [62], 2023	Low/Prospective observational study	14	180 days	Pain (BPI/NRS, change from baseline); worst/least pain; function/QoL; safety.	Mean pain=−2.9 (95 % CI [−4.6, −1.2], P=0.0028). Worst pain: −3.5 (P=0.0078). Least pain: −2.9 (P=0.0010). Function/QoL: −20.7 (95 % CI [−30.7, −10.7], P=0.0007). Safety: No AEs reported.	Small sample size; short follow-up; single-center; no control group; target limited to ICBN predominant pain.
	DesRochers et al. [63], 2024	Low/systematic review	124	Varied across studies	Pain scores (VAS/NRS)	Pooled effect (random-effects, Hedges' g): −0.963 (95 % CI [−1.55, −0.373]; z = −3.20), indicating a moderate-to-large reduction in pain. Between-study heterogeneity: I^2=70.1 %.	High heterogeneity; limited total evidence base; risk of bias; long-term safety data sparse.

Table 3: (continued)

Treatment strategies	Reference	Evidence level/ Design	Patients	Follow-up	Outcomes variables	Main results	Study limitations
SCS	Li et al. [64], 2025	Low/Retrospective observational study	78	6 months	Pain relief (VAS); predictors of response.	78.2 % met the ≥4-point VAS reduction. VAS decreased from 7.87 ± 0.79 to 3.72 ± 1.32 (mean change=−4.15). Younger age: HR=0.882 (95 % CI [0.802, 0.970], P=0.009). Lower BMI: HR=0.659 (95 % CI [0.487, 0.891], P=0.007). Shorter pain duration: HR=0.696 (95 % CI [0.505, 0.959], P=0.027). Psychological treatment: HR=0.015 (95 % CI [0.001, 0.377], P=0.011). Prior radiotherapy: HR=2.139 (95 % CI [1.219, 5.808], P=0.029). AUC=0.927.	Retrospective, observational design; small, single-center sample; short follow-up; safety under-reporting.
PENS	Wong et al. [65], 2024	Very low/Case report	1	6 months	Pain intensity (NRS); function (ROM); durability/retreatment; safety.	Pain intensity: Decreased from 7/10 to 3/10. Full ROM of the left arm regained. Pain relief was recaptured after a repeat serratus plane PENS performed about 16 months later. No immediate complications.	Single-patient case report with no control and potential concurrent or preceding interventions
rTMS	Kataria et al. [66], 2024	Moderate/RCT	34	About 3 weeks	Pain outcomes (VAS, SF-MPQ); QoL (FACT-B); thermal sensitivity (QST); safety.	rTMS reduced VAS and SF-MPQ (P<0.0001 for both). QoL: Total score increased with rTMS (P=0.0006); QST: CDT (P=0.0001), WDT (P=0.0033), CPT (P=0.0078), HPTT (P=0.0078), HPT (P=0.4407), and CPTT (P=0.7500) were not significant. The proportion of patients displaying side effects was low (P>0.05).	Short follow-up; small, single-center cohort; no advanced neuroimaging or modeling to confirm cortical mechanisms.
AFG	Caviggioli et al. 2011	Low/Prospective comparative cohort	113	Mean 13 months	Pain intensity (VAS); analgesic consumption; safety.	Mean VAS decreased 3.23 ± 2.96 points in AFG vs. 1.04 ± 2.71 in control, P=0.0005. Among 34 treated patients with medication records, 28 discontinued analgesics. Authors describe AFG as safe, relatively non-invasive, and rapid.	Non-randomized, single-center comparison; pain outcome measured by VAS only; short follow-up.
	Maione et al. [70], 2014	Low/Prospective comparative cohort	96	Mean 10 months	Pain intensity (VAS)	Mean VAS decrease of 3.1 (SD=2.7) in AFG vs. 0.9 (SD=2.6) in control, between-group mean difference 2.2, P<0.005.	Non-randomized, single-center study, pain outcome measured by VAS only; short follow-up.

Table 3: (continued)

Treatment strategies	Reference	Evidence level/ Design	Patients	Follow-up	Outcomes variables	Main results	Study limitations
	Lisa et al. [71], 2020	Low/Prospective interventional study	37	6 months	Pain (VAS); scar quality (POSAS); predictors	VAS: From baseline 6.9 (SD 1.3) to −0.39 (SD 2.09). POSAS: Significant improvements in both observer and patient scales (P<0.0001). Predictors: Smoking and axillary dissection predicted reduced clinical efficacy (P=0.0227, P=0.0066).	Strongly biased by the limited number of patients; short follow-up.
	Chen et al. [67], 2024	Low to Moderate/Meta-analysis	812	Varied across studies	Pooled effect on pain (VAS); safety.	VAS: Mean difference=−2.17 (95 % CI [−2.95, −1.39]), I^2=59 %, P<0.00001, favoring AFG. Complications: Capsular contracture (1.4 %), seroma (0.5 %), hematoma (0.5 %), implant infection or dehiscence (0.5 %); implant rupture (0.2 %); revision surgery: 4.1 %.	Heterogeneity in outcome reporting and timepoints; many studies are ineligible for meta-analysis due to mixed scales or lack of controls.
TMR	O'Brien et al. [72], 2020	Low/Preliminary cohort	11	Mean 5.1 months	QOL (BREAST-Q PWBC); safety.	QOL: Mean 77.5/100 (individual 85, 85, 80, 60). No 30-day readmissions. No minor or major complications. One patient developed a superficial wound related to subsequent radiation therapy at 5 months.	Small sample, no control group; short follow-up; lack of a specific PMPS diagnostic/measurement protocol.
RPNI	Hart et al. [74], 2022	Very low/Case series	7	Median 4 months	Pain outcome (VAS); safety.	Median VAS decreased from 9 to 1 (P=0.02). 86 % reported pain-free or "considerable improvement". 1 cellulitis managed with oral antibiotics.	Small single-surgeon, retrospective series; short follow-up; no control group.
	Nolan et al. [75], 2025	Very low/Proof-of-concept study	12	Median 19 months	Incidence of intercostal neuroma-related PBSPS; safety.	1/19 breasts (5.3 %); 1/12 patients (8.3 %) developed PBSPS. Survey response 75 %. Complications (per breast): Seroma (31.6 %), surgical-site cellulitis (5.3 %).	Small, single-surgeon, retrospective, no control group; possible recall or missing-data bias.

GSNP, grading system for neuropathic pain; CGIC, clinical global impression of change; PNP, peripheral neuropathic pain; NPRS, numeric pain rating scale; PRO, patient-reported outcome; PGIC, patient global impression of change; QOLS, quality of life scale; GPE, global perceived effect; POD, postoperative day; BPI, brief pain inventory; HR, hazard ratio; AUC, area under the (ROC) curve; ROM, range of motion; FACT-B, functional assessment of cancer therapy – breast; QST, quantitative sensory testing; CDT, cold detection threshold; WDT, warm detection threshold; CPT, cold pain threshold; HPTT, heat pain tolerance threshold; HPT, heat pain threshold; CPTT, cold pain tolerance threshold; SD, standard deviation; PBSPS, post-breast surgery pain syndrome.

modest benefit, whereas HCCP shows more consistent local analgesia. Nerve blocks are best used for diagnosis and stratification, guiding escalation to neuromodulation. For refractory PMPS, interventional or ablative approaches can be considered, although evidence and protocols vary. AFG offers both scar remodeling and pain reduction with generally low reported complication rates, while TMR and RPNI target neuroma or nerve-injury phenotypes with promising but preliminary data. For ease of comparison, Table 3 summarizes research in the treatment section. Adjunctive exercise programs, psychological therapies, and laser acupuncture may further improve function and well-being. To operationalize the above components in clinical practice, we propose an integrated clinical pathway that links risk factor identification, perioperative prevention, and multimodal management of PMPS (Figure 3).

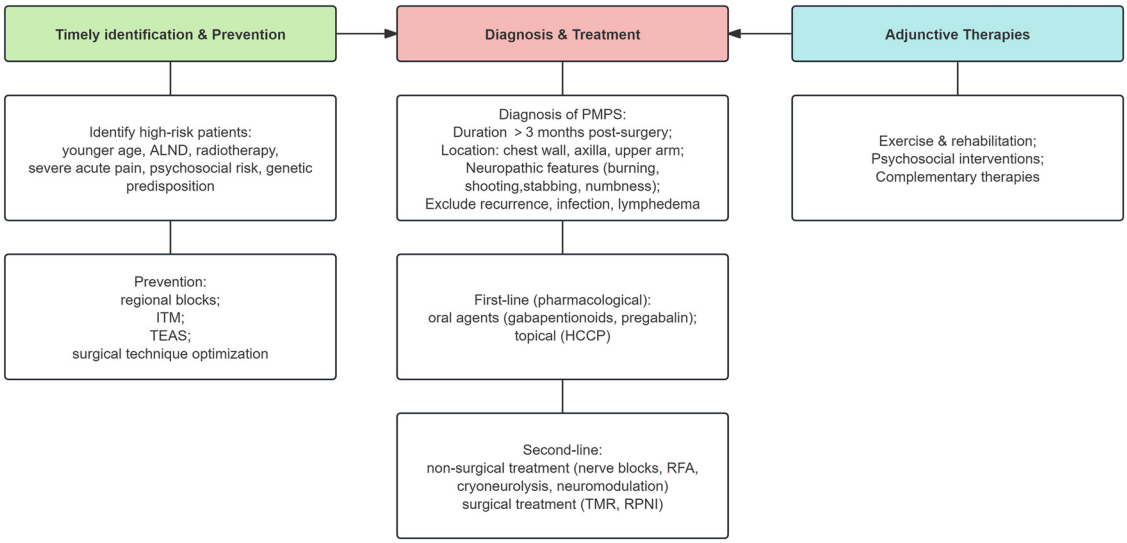

Figure 3: Integrated clinical pathway for prevention, diagnosis, and management of post-mastectomy pain syndrome (PMPS).

Conclusions and outlook

PMPS remains a prevalent and undertreated complication that significantly impairs QOL in breast cancer survivors. Current evidence underscores the multifactorial etiology of PMPS, involving surgical injury, adjuvant therapy, inadequate acute pain control, as well as psychosocial and genetic factors. Advances in diagnostic criteria and epidemiological studies have improved recognition, yet wide gaps persist between disease burden and effective management. Although pharmacological and topical therapies offer symptomatic relief, their efficacy is often modest and limited by tolerability. Interventional approaches, including nerve blocks, radiofrequency ablation, cryoneurolysis, and neuromodulation, provide promising results in selected patients, while surgical innovations such as autologous fat grafting and nerve reconstruction have opened new avenues for refractory cases. However, adoption varies across healthcare settings due to resource needs, workforce training, and device costs. Nerve blocks are most scalable where ultrasound is available and anesthesiologists are appropriately trained. Radiofrequency ablation and cryoneurolysis require dedicated generators and probes, which can limit uptake. Neuromodulation and nerve reconstruction procedures need higher technical demands. Given these constraints, clinicians should actively identify risk factors early and prioritize low-cost, low-risk measures first, and then escalate to equipment or expertise-intensive interventions for refractory cases, particularly in low- and middle-income settings.

In the future, while surgical advances continue to improve survival outcomes for breast cancer patients, equal attention should be directed toward enhancing their QOL. Patients suffering from chronic postoperative pain deserve timely and appropriate interventions. Accordingly, research on PMPS is not only clinically meaningful but also increasingly necessary. Future efforts should focus on several aspects. First, standardized diagnostic frameworks and validated assessment tools should be integrated into routine PMPS follow-up, enabling early identification and timely intervention. Second, future clinical trials must emphasize robust methodology, larger cohorts, and long-term outcomes to define the comparative effectiveness of existing and emerging therapies. Third, interdisciplinary strategies that combine pharmacological, interventional, surgical, and psychosocial modalities are likely to yield the greatest benefit. Finally, individualized PMPS management can align patient-reported pain location and extent with ultrasound-guided targeting to deliver precise nerve blocks, neuromodulation, or perioperative nerve reconstruction. As evidence grows, genetic profiling may further refine risk stratification and prevention. By aligning epidemiological insights with innovative therapeutic strategies, clinicians can move closer to bridging the gap between recognition and effective treatment, ultimately improving survivorship care for breast cancer patients.

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